

Environmental Monitoring of a Dry-Blend Powder Line

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Abstract. We review twelve months of environmental swabs from a dry-blend powder line. Zone 1 contact surfaces stayed below action limits in every month, while Zone 3 floor drains drove most presumptive positives. Sanitation after wet cleaning, not the blend recipe, explained the seasonal peak. We propose a risk-weighted sampling rotation and report the effect of the change on detection rates.

1 Introduction

Dry powder facilities depend on keeping water out. When water enters during a wet clean, organisms that persist in dry niches can grow, and a single harborage site can seed a whole zone³. Environmental monitoring (EM) programs therefore sample by zone, from product-contact surfaces (Zone 1) outward to non-production areas (Zone 4).

This report has three aims:

- Describe positives by zone, site type and month.
- Test whether wet cleaning predicts the next positive.
- Propose a rotation that weights high-risk sites.

2 Methods

2.1 Sampling

Sites were swabbed with pre-moistened sponges over 100 cm² where the surface allowed, otherwise the whole accessible surface. Each sample was enriched and screened for *Salmonella* spp. and *Cronobacter* spp. Presumptive positives were confirmed by the contract laboratory.

2.2 Risk score

Each site received a risk score from its zone weight w_z , its moisture exposure m and its history of positives h :

$$R = w_z \cdot (1 + m) + 2h$$

Sites with $R \geq 6$ were sampled weekly; the rest rotated monthly. A sampling plan of 1. weekly, 2. monthly and 3. quarterly tiers was kept.

1. Weekly: all Zone 1 sites and any site with $R \geq 6$.
2. Monthly: remaining Zone 2 and Zone 3 sites.
3. Quarterly: Zone 4.

Table 1. Microbiological limits by product stage (n, c, m, M sampling plan)

Product stage	Organism	Sampling plan				Action
		n	c	m (CFU/g)	M (CFU/g)	
Base powder	Aerobic plate count	5	2	1,000	10,000	Hold lot
	<i>Enterobacteriaceae</i>	10	2	0	10	Hold lot
	<i>Cronobacter</i> spp.	30	0	Absent in 10 g		Reject
Finished blend	<i>Salmonella</i> spp.	60	0	Absent in 25 g		Reject
	Yeast and mould	5	2	50	500	Investigate

n = samples per lot; c = samples allowed between m and M. Limits follow the customer specification.

3 Results

Across 1,284 swabs, 41 were presumptive positive and 17 confirmed. Zone 3 drains accounted for 11 of the 17 confirmed results. No Zone 1 site was

confirmed positive. The monthly pattern (Figure 1) peaks in March and August, the two months with the most wet cleans.

Confirmed positives followed a wet clean within seven days in 13 of 17 cases (76%). A post-clean

dry-out step, added in May, reduced that share to 2 of 6 in the second half of the year.

Table 2. Swab results by zone, first and second half of the year

Zone	Swabs H1	Pos. H1	Rate H1 (%)	Swabs H2	Pos. H2	Rate H2 (%)
Zone 1	312	0	0.0	318	0	0.0
Zone 2	204	3	1.5	198	1	0.5
Zone 3	96	9	9.4	101	2	2.0
Zone 4	27	1	3.7	28	1	3.6
Total	639	13	2.0	645	4	0.6

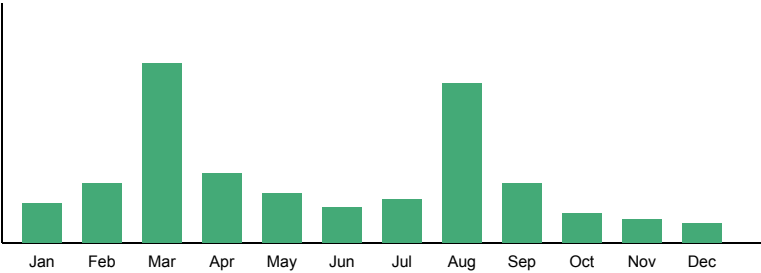


Figure 1. Presumptive positives per month. Peaks follow wet-clean months.

4 Discussion

The data point to water, not the blend recipe, as the driver. Drains and the floor around them stay wet for hours after a clean, and traffic carries that water toward Zone 2. The dry-out step shortened that window. A risk-weighted rotation puts more swabs where water lingers without adding total samples.

Two limits apply. Twelve months is short for seasonal claims, and the contract laboratory changed its

enrichment method in October, which may shift detection.

5 Conclusion

Keep Zone 1 weekly. Move drains to weekly after any wet clean. Keep the dry-out step. Review the rotation after the next twelve months.

³ A harborage site is a niche where organisms survive cleaning, such as a cracked seal or a hollow roller.

Appendix A. Site register (excerpt)

Site ID	Zone	Description	Surface	R	Frequency
Z1-014	1	Blender discharge chute, inner wall	Stainless steel	7.5	Weekly
Z1-022	1	Filler hopper lid gasket	Silicone	8.0	Weekly
Z2-031	2	Filler frame, underside of cross-member near the lift motor	Painted steel	4.5	Monthly
Z3-007	3	Floor drain D-3, grate and basket	Cast iron	9.0	Weekly after wet clean
Z3-011	3	Wash-down hose reel and nozzle holster	Plastic	6.5	Weekly
Z4-002	4	Corridor door threshold to warehouse	Rubber seal	2.0	Quarterly